

The Universe Has a Resolution Limit: New Theory Derives the Exact Mass Where Quantum Mechanics Ends

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CALABASAS, Calif., Feb. 9, 2026 /PRNewswire/ -- A new theoretical framework released today proposes that the vacuum of space is not a smooth void, but a geometrically structured medium with a finite information density. The **Selection-Stitch Model (SSM)**, developed by Raghu Kulkarni, CEO of IDrive Inc. and independent researcher, offers exact derived values for two of physics' most elusive numbers: the effective "pixel size" of spacetime and the precise mass limit where quantum superposition fails.

These findings, detailed in two papers published on Zenodo, provide a theoretical map for the current experimental race to test quantum gravity, recently highlighted in the journal *Nature*.

The Resolution of Reality (0.77 Planck Lengths)

For a century, physicists have assumed the smallest unit of distance is the Planck Length. However, standard theory treats this as an abstract limit without describing how spacetime is structured at that scale.

Kulkarni's research suggests that quantum information cannot be packed continuously. Instead, the informational capacity of the vacuum behaves mathematically like a Face-Centered Cubic (FCC) lattice—nature's most efficient packing algorithm. By calculating the density of this informational structure, the **Selection-Stitch Model (SSM)** derives a new fundamental constant:

The Geometric Vacuum Constant (approximately 0.77 times the Planck Length).

"We have long treated the Planck scale like a blurry limit," said Kulkarni. "But if you treat space as an information storage medium, geometry dictates a specific packing efficiency. The universe has a specific resolution, and it is tighter than the standard Planck length suggests."

The "Geometric Resolution Limit" of Quantum Mechanics

The SSM also addresses the "measurement problem": why do subatomic particles behave like waves (superposition), while macroscopic objects do not?

The new theory proposes that this is a **"Geometric Resolution Limit."** In quantum mechanics, mass and wavelength are inversely related—heavier objects have smaller wavelengths. Kulkarni proposes that when an object's mass exceeds a specific threshold, its wavelength becomes smaller than the "pixel size" of the vacuum itself. The vacuum can no longer resolve the wave, forcing it to collapse into a classical state.

Kulkarni's calculations place this saturation point—the "Mass-Decoherence Limit"—at approximately **28 micrograms**. Objects heavier than this limit exceed the vacuum's resolution and are forced to behave classically.

Convergence with Roger Penrose

Significantly, this derived limit of **28 micrograms** converges remarkably with the prediction of Nobel Laureate **Roger Penrose**. Penrose's "Gravitational Objective Reduction" model predicts quantum collapse near the Planck Mass (approximately 21.7 micrograms) due to the instability of spacetime curvature.

"While Penrose arrived at this number via General Relativity, we arrived at nearly the same number via pure lattice geometry," Kulkarni noted. "The fact that two completely different theoretical approaches converge on the same 'mass cliff' suggests that this limit is a fundamental physical boundary that experimentalists will soon encounter."

Connecting to Experimental Breakthroughs

This theoretical derivation arrives just as experimental physics is approaching these scales. A recent study highlighted in *Nature* ("Levitated Nanoparticles for Testing Quantum Gravity") detailed efforts to measure gravitational coupling in microscopic particles, inching closer to the transition between quantum and classical mechanics.

"Experimentalists are digging this tunnel from one side, building tinier and tinier scales," Kulkarni noted. "The **Selection-Stitch Model (SSM)** digs from the other side, providing the exact coordinates where the quantum world ends and gravity begins."

Availability

The full theoretical papers and the general model are available for open access:

- **General Theory Hub:** [Selection-Stitch Model \(SSM\) Theory](#)
- **Paper 1 (Vacuum Geometry):** [Geometric Renormalization of the Speed of Light and the Origin of the Planck Scale](#) (DOI: 10.5281/zenodo.18447672)
- **Paper 2 (Mass Limit):** [Discrete Wave Mechanics: Deriving the Mass-Decoherence Limit](#) (DOI: 10.5281/zenodo.18453492)

About the Selection-Stitch Model

The **Selection-Stitch Model (SSM)** is a candidate theory for Quantum Gravity that models spacetime as a discrete, chiral tensor network. It recovers General Relativity and Quantum Mechanics as emergent behaviors of a "stiff" geometric vacuum, resolving tensions in cosmology (Hubble Tension) and fundamental mass limits without invoking dark energy or arbitrary constants.

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